BASH Script Automation

Daniel Burger

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SVAD111

Ivy Tech Community College

Professor Fyfe

1. Bash Script Purpose

The purpose of this script is to automatically create a new folder and copy a text file from the myscripts folder into it. This helps a user organize files without having to type multiple commands every time.

Using a script is helpful because it saves time, reduces mistakes, and makes sure the same steps are done the same way each time. Instead of manually creating folders and copying files, the script handles everything in one run.

2. Script Creation

To create the script, you would open a text editor in Linux such as nano or vim and create a new file named something like file_copy.sh. The ".sh" extension is used so it is clear that the file is a Bash script.

For the purpose of this script, we will be using three simple lines (not counting documentation):

#!/bin/bash – a "shebang" which tells Linux how to execute the program,

mkdir – "make directory" to create a folder, and

cp – "copy" to copy files from the old folder to the new one.

After writing the script you would save and exit the file and then use the "chmod" command to give it permission to run as a program. This is done by changing the file's permissions, so the system knows it is executable. Without this step, the script would just be a text file and would not run.

3. Folder and File Operations

The script performs two main tasks:

First, it creates a new folder using "mkdir". This folder acts as the destination where the file will be stored. Creating the folder first is important because the file needs a place to go.

Second, the script copies a text file from the myscripts folder into the new folder, using "cp". This keeps the original file unchanged while also placing a copy in the new location for organization or backup purposes.

Together, these commands make sure the folder exists before trying to copy the file into it.

4. Script Structure

A Bash script is just a list of Linux commands written in order. When the script runs, it goes through each command one at a time from top to bottom.

In this case, the script first creates a folder and then copies a file into that folder. Each step depends on the one before it, so the order matters.

This simple structure makes scripts easy to understand and reuse.

5. Users and Groups Considerations

Only certain users and groups should be able to run or modify the script.

The instructor or owner of the script would have full control over it, meaning they can edit and run it. A group such as instructors or administrators could be allowed to run the script but not change it. Other users, such as students, would usually not have access unless it is specifically needed.

Permissions would be set so that only the correct users can execute or modify the script. This helps keep the system secure and prevents accidental changes.

6. Script Documentation

Script name: file_copy.sh

```
#!/bin/bash

# Create destination folder
mkdir project_backup

# Copy file from myscripts folder into new directory
cp myscripts/sample.txt project_backup/
```

The script contains simple commands that create a folder and copy a file from myscripts into that folder.

Documenting scripts is important because it helps other people understand what the script does. It also makes it easier to fix or update later, especially if someone else needs to use it. Good documentation helps keep things organized and avoids confusion in the future.